Lecture 3

Transmission Modes

Introduction

- The transmission mode is the process of transferring data and information between two devices.
- The transmission mode is also known as the communication mode.
- Each communication channel has a direction associated with it, and transmission media provide the direction. Therefore, the transmission mode is also known as a directional mode.
- The transmission mode is defined in the physical layer.

Figure *Transmission medium and physical layer*

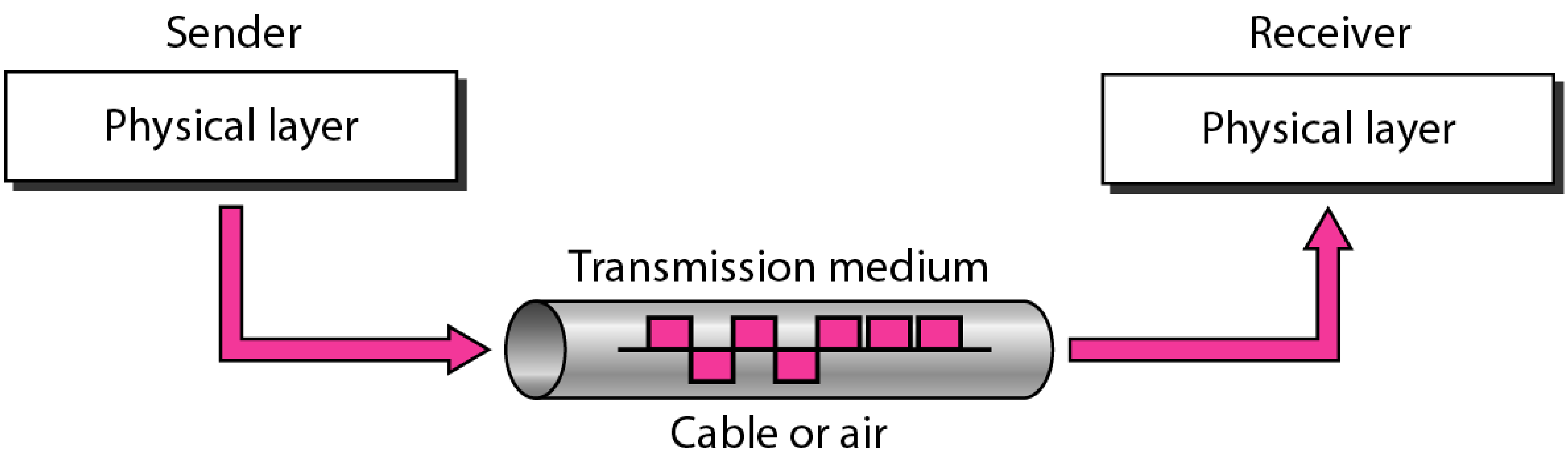
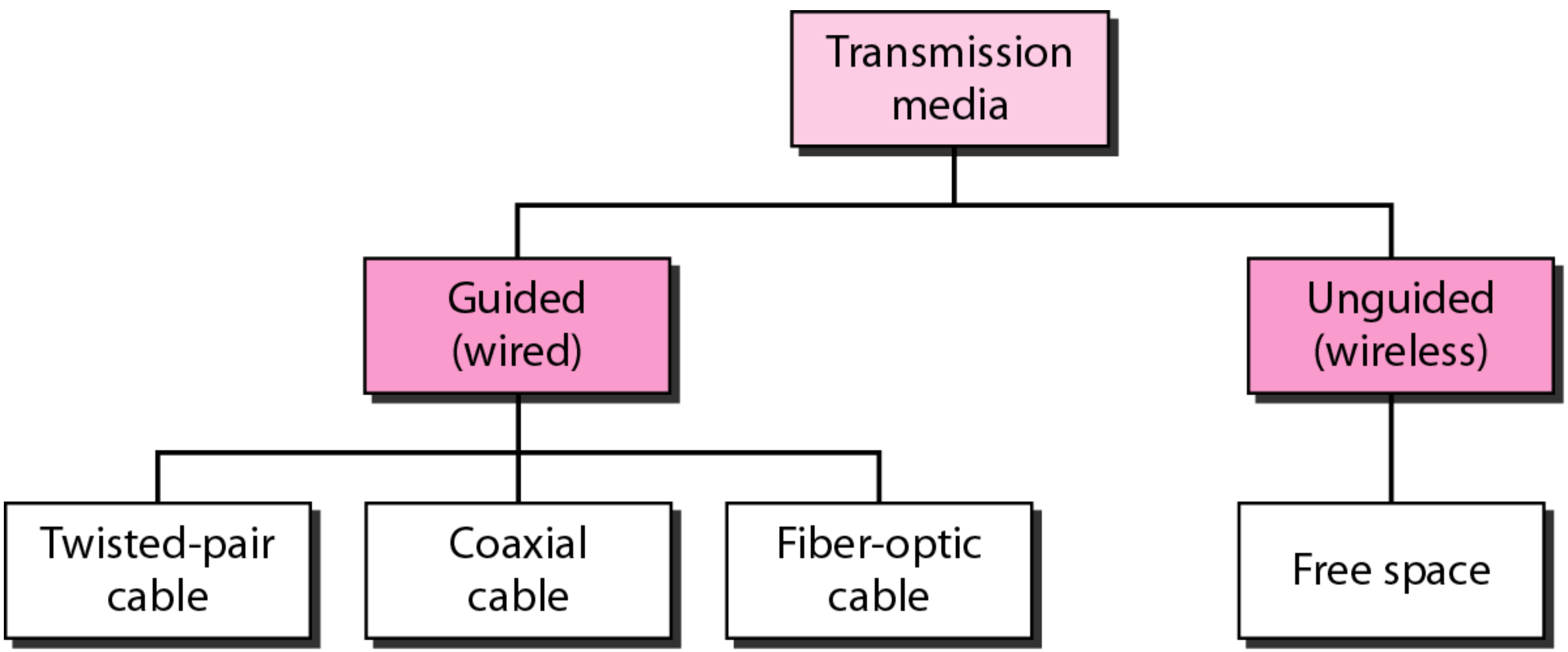


Figure *Classes of transmission media*



Guided Media: Wired

- Wired media allows signal energy enclosed and guided within a solid medium.
- This media is used either for point-to-point links or a shared link with various connections, include twisted-pair cable, coaxial cable, and fiber-optic cable.

Guided Media

- **Twisted Pair Cable**: It made up of insulated copper wires twisted together that's why it called is as twisted pair cable. Mostly used in telephone networks and Ethernet cables. It is affordable.
- **Coaxial Cable**: In this cable one central conductor surrounded by insulation, a metallic shield, and an exterior cover. This type of cable used in television networks and for long-distance communication lines. It provides better protection against interruption than twisted pair cables.
- **Optical Fiber**: Uses light signals to transfer data. Made up of glass or plastic fibers, it provides very high bandwidth and low signal reduction, making it ideal for long-distance and high-speed data transmission. It's immune to electromagnetic obstruction but more expensive than copper cables

Unguided Media: Wireless

- In the unguided media, the signal energy propagates through a wireless medium.
- Wireless media is used for radio broadcasting in all directions.
- This type of communication is often referred to as wireless communication.

Unguided Media: Wireless

- **Radio Waves:** Are used for wireless communication networks over large range distances, such as in broadcasting, cellular networks, and Wi-Fi. It can travel through walls and have a wide coverage area.
- **Microwaves:** It provide high-frequency radio waves used for one-to-one communication. Used in satellite and terrestrial communication.
- **Infrared:** Uses light signals just below the visible spectrum. Infrared is used for nearby communication, such as remote controls and short-range data transfer (e.g., between a laptop and a mobile device).
- **Satellite Communication:** Satellite communication used for broadcasting, GPS, and global communications.

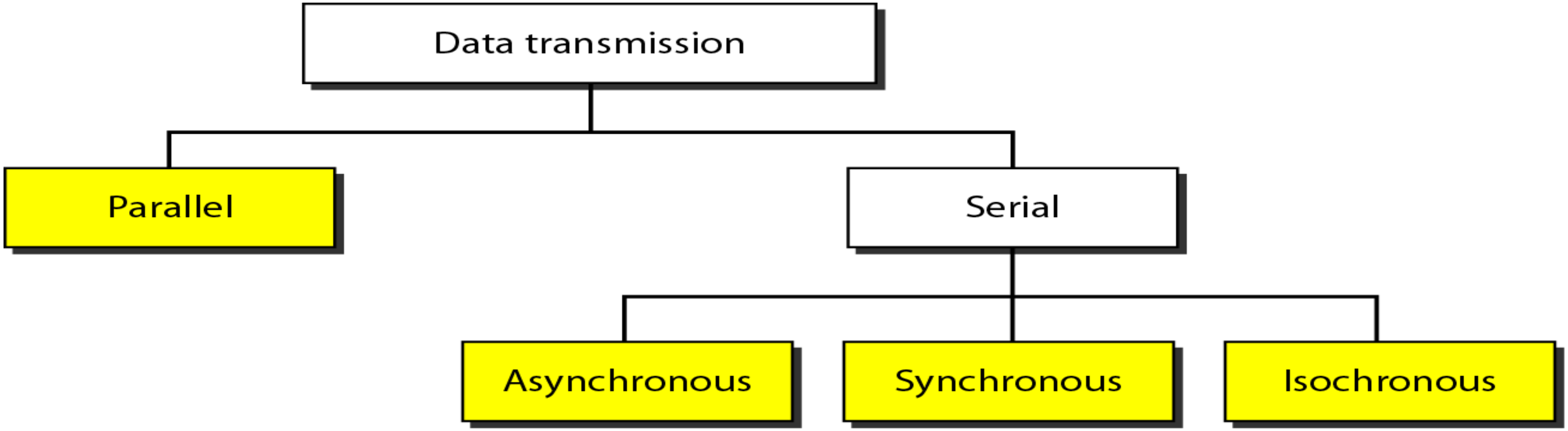
Data Transmission Modes

- The transmission of binary data across a link can be accomplished in either parallel or serial mode.

In parallel mode, multiple bits are sent with each clock tick.

In serial mode, 1 bit is sent with each clock tick. While there is only one way to send parallel data, there are three subclasses of serial transmission: asynchronous, synchronous, and isochronous.

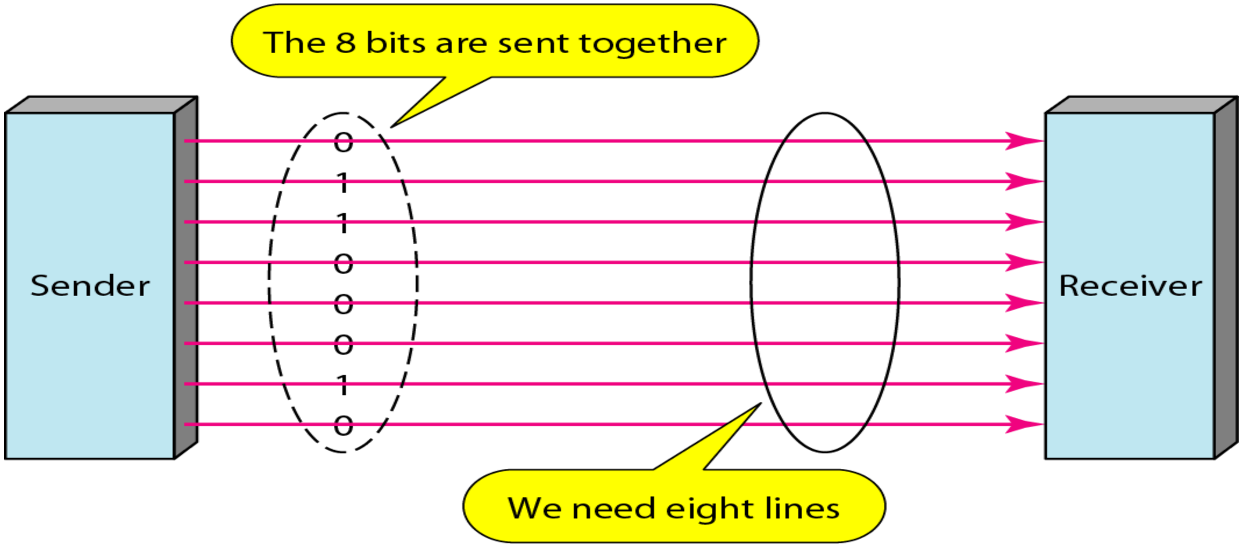
Data Transmission and Modes



Parallel transmission

Example

Mechanism: Use *n* wires
to send *n* bits at one time. That way each bit has its own wire, and all *n* bits of one group can be transmitted with each clock tick from one device to another.



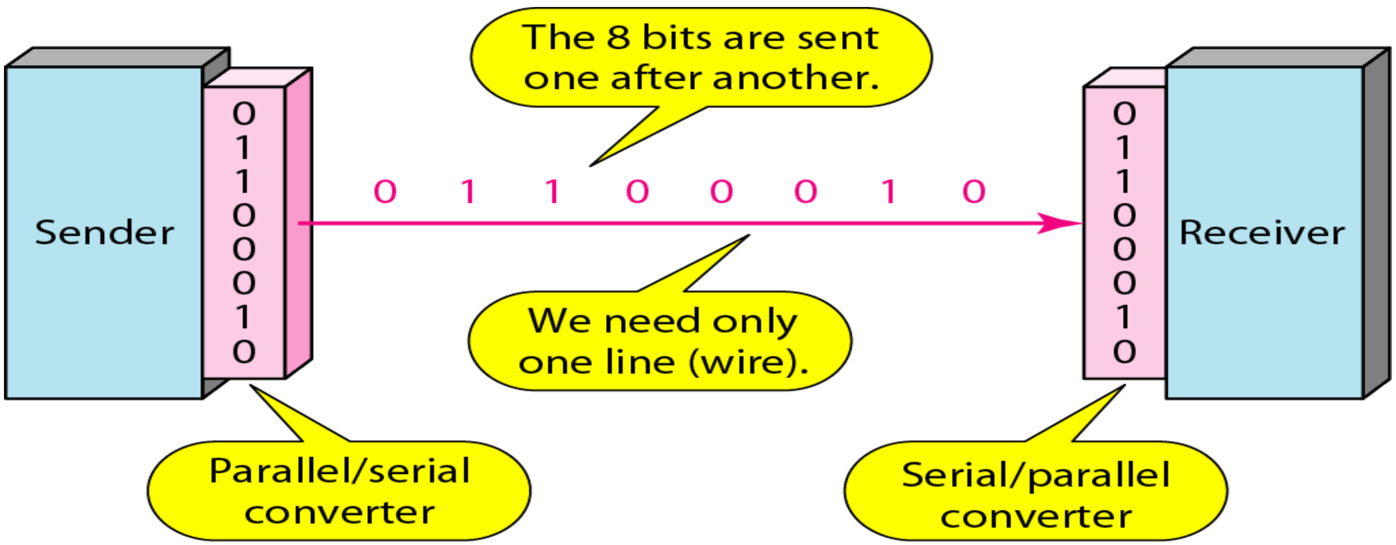
The advantage of parallel transmission is speed. All else being equal, parallel transmission can increase the transfer speed by a factor of n over serial transmission.

But there is a significant disadvantage: cost. Parallel transmission requires n communication lines (wires in the example) just to transmit the data stream. Because this is expensive, parallel transmission is usually limited to short distances.

Serial transmission

Example

Mechanism: In serial transmission one bit follows another. Therefore only one communication channel rather than *n* to transmit data between two communicating devices



The advantage of serial over parallel transmission is that with only one communication channel, serial transmission reduces the cost of transmission over parallel by roughly a factor of n.

Timing of Serial Transmission

- Serial transmission mechanisms can be divided into three broad categories (depending on how transmissions are spaced in time):
  - **Asynchronous transmission** can occur at any time with an arbitrary delay between the transmission of two data items
  - **Synchronous transmission** occurs continuously with no gap between the transmission of two data items
  - **Isochronous transmission** occurs at regular intervals with a fixed gap between the transmission of two data items

Note

In asynchronous transmission, data bits sent start bit (0) at the beginning and stop bits (1) at the end of each byte. There may be a gap between each byte.

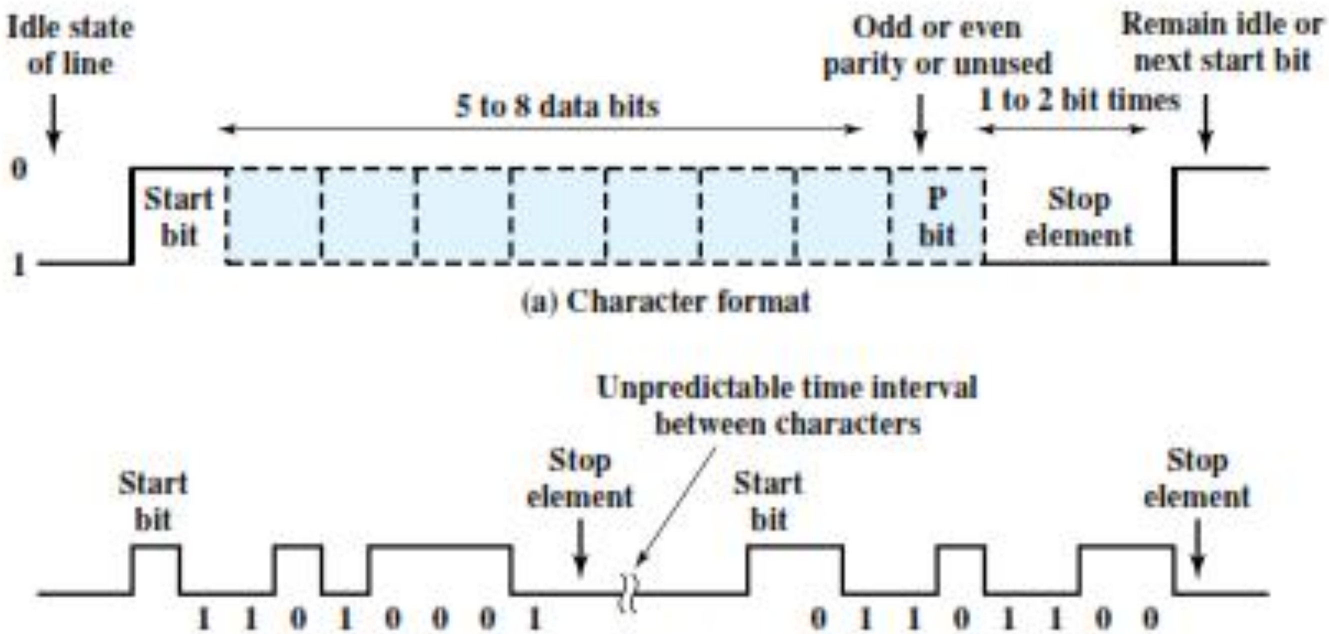
ASYNCHRONOUS TRANSMISSION

- Asynchronous transmission is so named because the timing of a signal is unimportant
- Information is received and translated by agreed upon patterns
- Avoids the timing problem by not sending long, uninterrupted streams of bits.
- Data are transmitted one character at a time, where each character is five to eight bits in length.
- Timing or synchronization must only be maintained within each character; the receiver has the opportunity to resynchronize at the beginning of each new character.

ASYNCHRONOUS TRANSMISSION

- The reception of digital data involves sampling the incoming signal once per bit time to determine the binary value.
- Transmission Impairments corrupts the signal, bringing about errors and difficulty in timing.
- Timing problems require a mechanism to synchronise a transmitter and a receiver

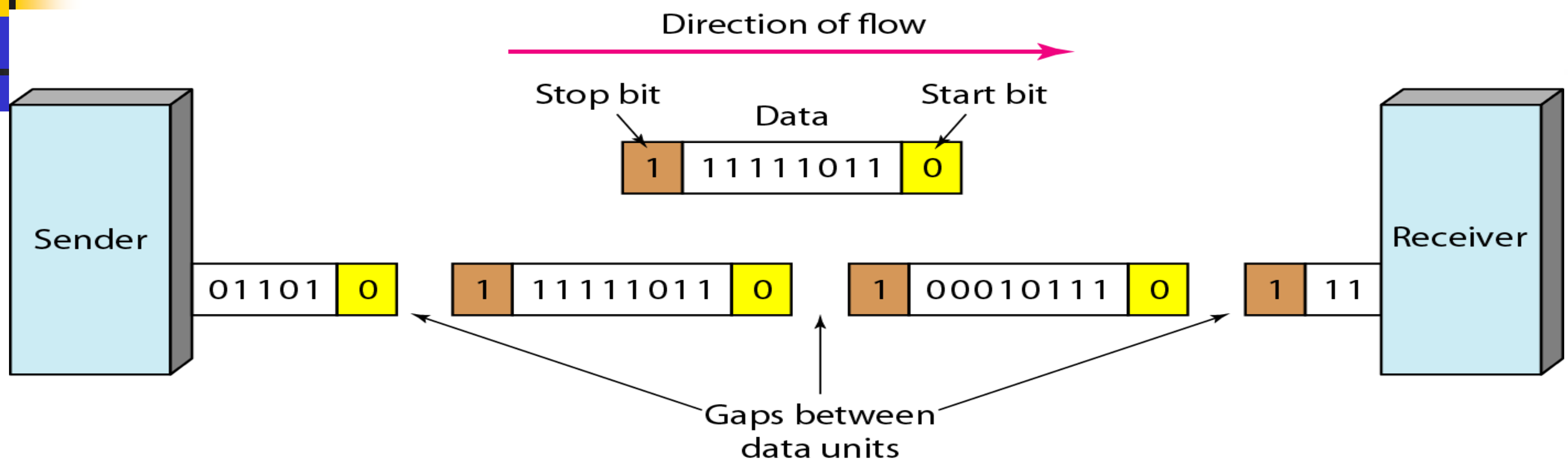
ASYNCHRONOUS TRANSMISSION



(a) Character format

(b) 8-bit asynchronous character stream

Example: Asynchronous transmission



- The addition of stop and start bits and the insertion of gaps into the bit stream make asynchronous transmission slower than forms of transmission that can operate without the addition of control information. But it is cheap and effective, two advantages that make it an attractive choice for situations such as low-speed communication. For example, the connection of a keyboard to a computer is a natural application for asynchronous transmission.
- A user types only one character at a time, types extremely slowly in data processing terms, and leaves unpredictable gaps of time between each character.
- Asynchronous data transfer systems usually have an error detection mechanism. If an error is detected the data can be resent.

Note

In synchronous transmission, data bits sent one after another without start or stop bits or gaps. It is the responsibility of the receiver to group the bits.

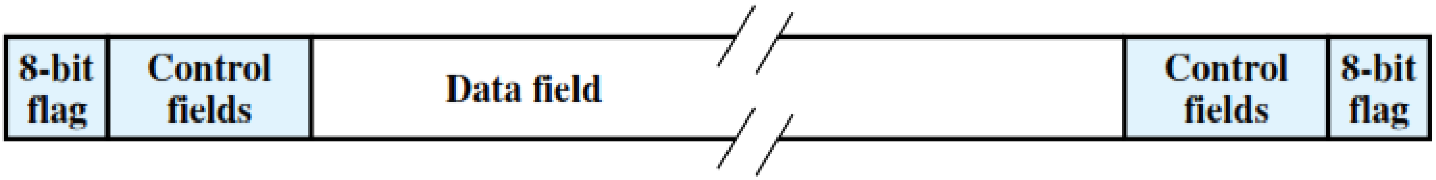
SYNCHRONOUS TRANSMISSION

- A block of bits is transmitted in a steady stream without start and stop codes. The block may be many bits in length.
- To prevent timing drift between transmitter and receiver, their clocks must somehow be synchronized.
- Can use either a separate clock line (This is good over short distance, because it's subject to impairments), or
- Embed a clock signal in data by proper encoding method (e.g. Manchester/Differential Manchester Encoding for digital signals and Use of the career frequency for analog signals.)

SYNCHRONOUS TRANSMISSION

- Another level of synchronization required, to allow the receiver to determine the beginning and end of a block of data.
- Each block begins with a *preamble bit pattern and generally ends* with a *postamble bit pattern.*
- Other bits are added to the block that convey control information used in the data link control procedure.
- The data plus preamble, postamble, and control information are called a **frame.**

Synchronous Frame Format

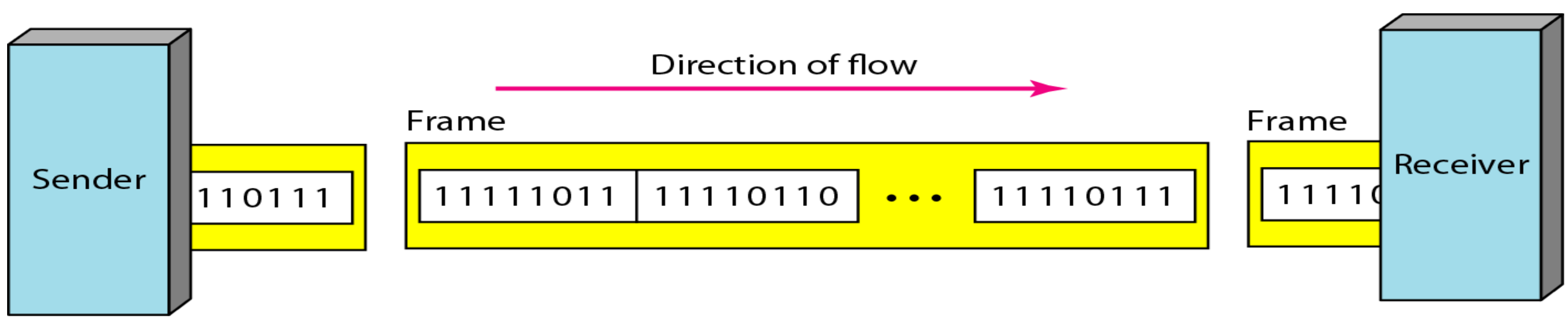


- The frame starts with a preamble called a flag, which is 8 bits long. The same flag is used as a postamble. The receiver looks for the occurrence of the flag pattern to signal the start of a frame
- This is followed by some number of control fields (containing data link control protocol information), then a data field (variable length for most protocols), more control fields, and finally the flag is repeated.

Timing becomes very important, therefore, because the accuracy of the received information is completely dependent on the ability of the receiving device to keep an accurate count of the bits as they come in.

The advantage of synchronous transmission is speed. With no extra bits or gaps to introduce at the sending end and remove at the receiving end, and, by extension, with fewer bits to move across the link, synchronous transmission is faster than asynchronous transmission.
For this reason, it is more useful for high-speed applications such as the transmission of data from one computer to another.
•Byte synchronization is accomplished in the data link layer.
•Synchronous data transfer systems usually have an error detection mechanism. If an error is detected the data can be resent.

Synchronous transmission



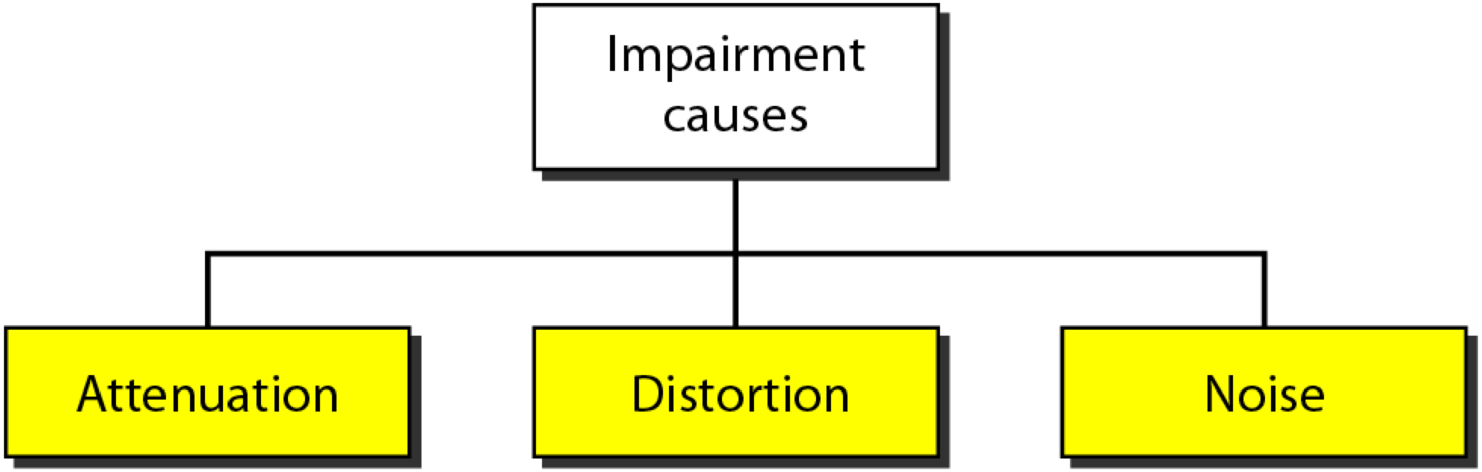
Isochronous Transmission

- It combine the concept of Asynchronous and Synchronous transmission
- In real-time audio and video, in which uneven delays between frames are not acceptable, synchronous transmission fails.
- The isochronous transmission guarantees that the data arrive at a fixed rate.
- Isochronous systems do not have an error detection mechanism (acknowledgment of receipt of packet) because if an error were detected, time constraints would make it impossible to resend the data.
- For example, TV images are broadcast at the rate of 30 images per second; they must be viewed at the same rate.
- If each image is sent by using one or more flames, there should be no delays between frames.
- For this type of application, synchronization between characters is not enough; the entire stream of bits must be synchronized.

Transmission Impairment

- Signals travel through transmission media, which are not perfect.
- The imperfection causes signal impairment.
- This means that the signal at the beginning of the medium is not the same as the signal at the end of the medium. What is sent is not what is received.
- Three causes of impairment are attenuation, distortion, and noise.

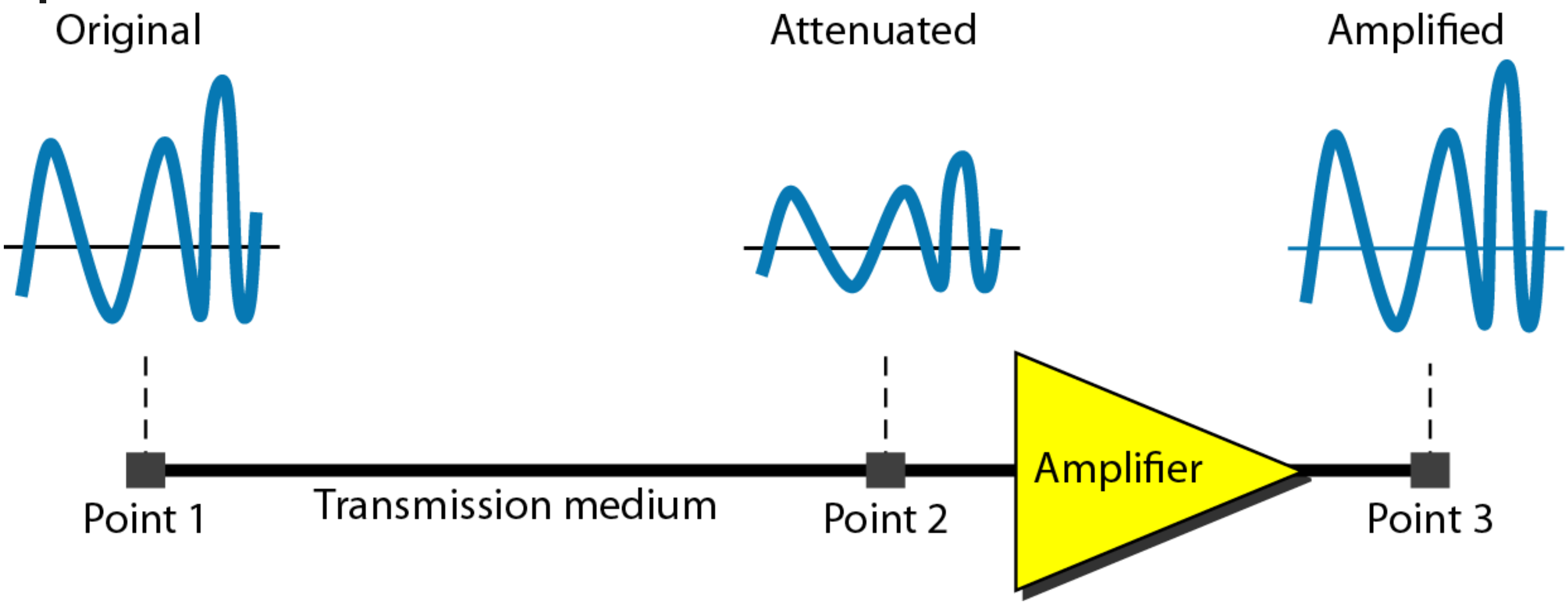
Figure *Causes of impairment*



Attenuation

- Attenuation means a loss of energy.
- When a signal, simple or composite, travels through a medium, it loses some of its energy in overcoming the resistance of the medium.
- To show that a signal has lost or gained strength, engineers use the unit of the decibel.
- The decibel (dB) measures the relative strengths of two signals or one signal at two different points.

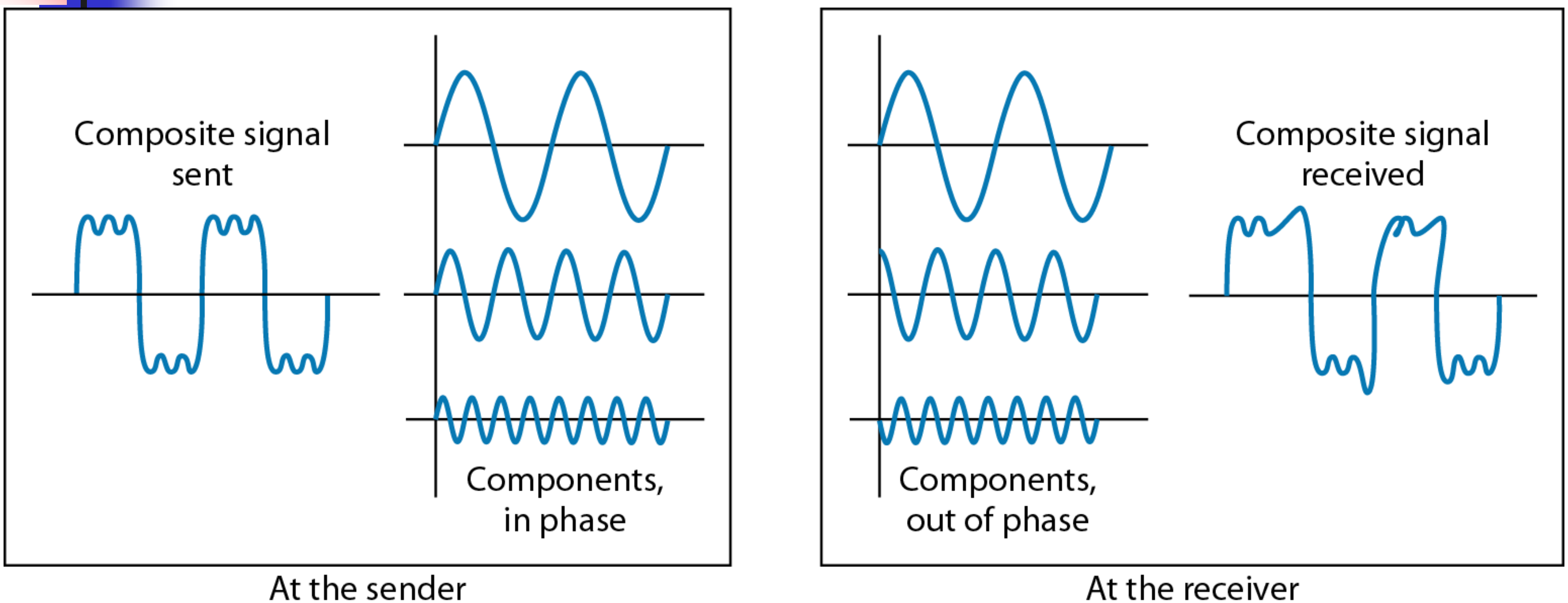
Figure *Attenuation*



Distortion

- Distortion means that the signal changes its form or shape.
- Distortion can occur in a composite signal made of different frequencies.
- As a result, signal components at the receiver have phases different from what they had at the sender.
- The shape of the composite signal is therefore not the same.

Figure *Distortion*



Noise

- Noise is another cause of impairment.
- Several types of noise, such as thermal noise, induced noise, crosstalk, and impulse noise, may corrupt the signal.
- **Thermal noise** is the random motion of electrons in a wire, which creates an extra signal not originally sent by the transmitter.
- **Induced noise** comes from sources such as motors and appliances. These devices act as a sending antenna, and the transmission medium acts as the receiving antenna.
- **Crosstalk** is the effect of one wire on the other. One wire acts as a sending antenna and the other as the receiving antenna.
- **Impulse noise** is a spike (a signal with high energy in a very short time) that comes from power lines, lightning, and so on.

Figure ***Noise***

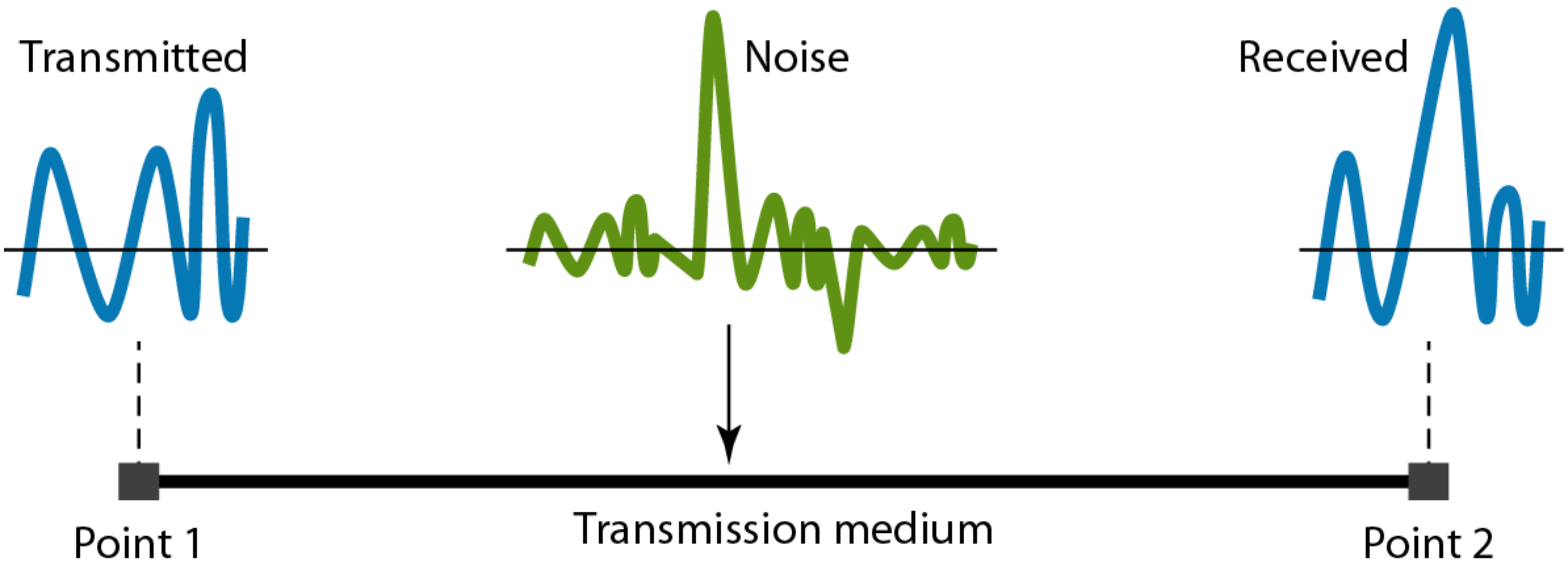
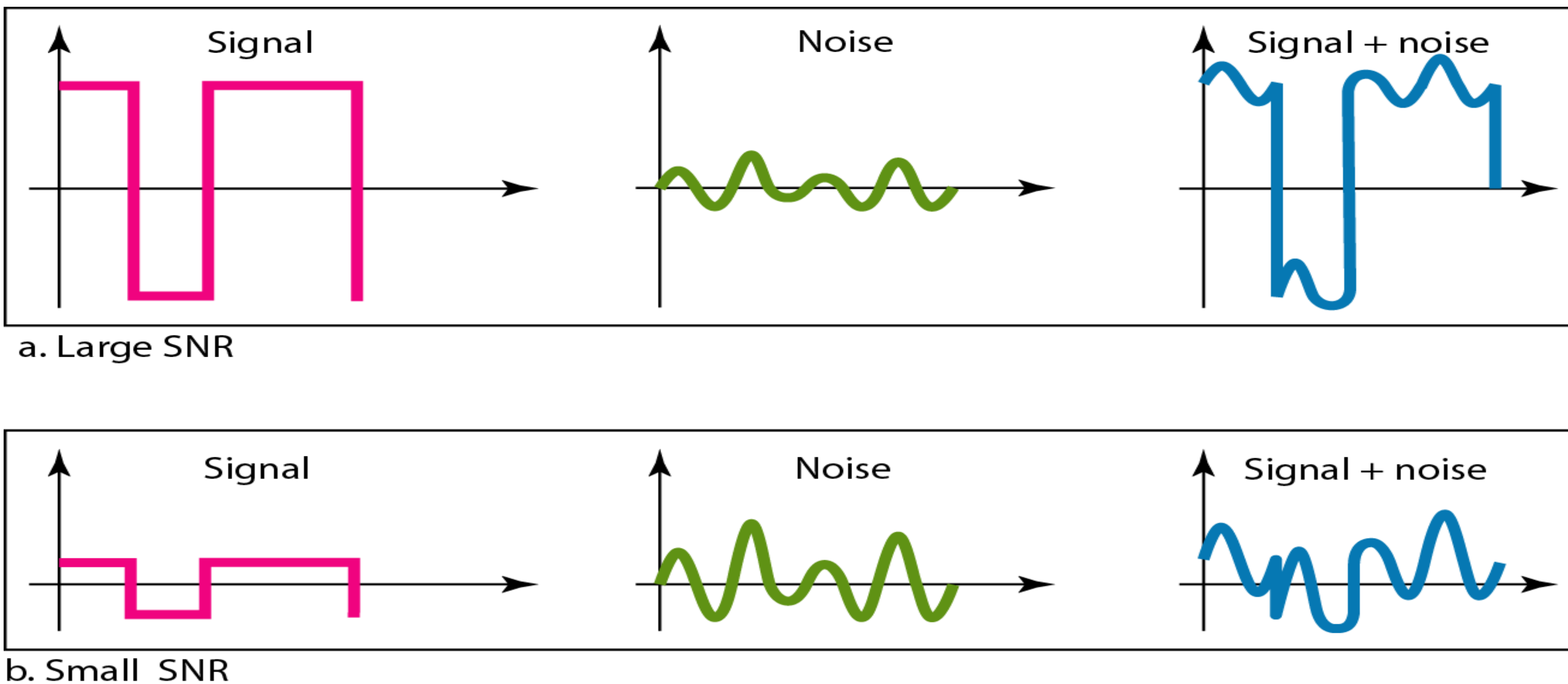


Figure ***Two cases of Signal to Noise Ratio: a high SNR and a low SNR***

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$



DATA RATE LIMITS

A very important consideration in data communications is how fast we can send data, in bits per second, over a channel. Data rate depends on three factors:

1. The bandwidth available
2. The level of the signals we use
3. The quality of the channel (the level of noise)

Note

Increasing the levels of a signal may reduce the reliability of the system.

Note

Increasing the levels of a signal may reduce the reliability of the system. Two theoretical formulas were developed to calculate the data rate: one by Nyquist for a noiseless channel, another by Shannon for a noisy channel.

Data Rate Calculation

Nyquist

Noiseless Channel

L = signal levels

$$\text{BitRate} = 2 \times \text{bandwidth} \times \log_2 L$$

Shannon

Noisy Channel

SNR = signal to noise ratio

$$\text{Capacity} = \text{bandwidth} \times \log_2(1 + \text{SNR})$$

Example 1

Consider a noiseless channel with a bandwidth of 3000 Hz transmitting a signal with two signal levels. The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Example 2

Consider the same noiseless channel transmitting a signal with four signal levels (for each level, we send 2 bits). The maximum bit rate can be calculated as

$$\text{BitRate} = 2 \times 3000 \times \log_2 4 = 12{,}000 \text{ bps}$$

Example 3

Consider an extremely noisy channel in which the value of the signal-to-noise ratio is almost zero. In other words, the noise is so strong that the signal is faint. For this channel the capacity C is calculated as

$$\boxed{C = B \log_2 (1 + \text{SNR}) = B \log_2 (1 + 0) = B \log_2 1 = B \times 0 = 0}$$

This means that the capacity of this channel is zero regardless of the bandwidth. In other words, we cannot receive any data through this channel.

Example 4

We can calculate the theoretical highest bit rate of a regular telephone line. A telephone line normally has a bandwidth of 3000. The signal-to-noise ratio is usually 3162. For this channel the capacity is calculated as

$$C = B \log_2 (1 + \text{SNR}) = 3000 \log_2 (1 + 3162) = 3000 \log_2 3163$$
$$= 3000 \times 11.62 = 34{,}860 \text{ bps}$$

This means that the highest bit rate for a telephone line is 34.860 kbps. If we want to send data faster than this, we can either increase the bandwidth of the line or improve the signal-to-noise ratio.

Example 5

We have a channel with a 1-MHz bandwidth. The SNR for this channel is 63. What are the appropriate bit rate and signal level?

Solution

First, we use the Shannon formula to find the upper limit.

$$\boxed{C = B \log_2 (1 + \text{SNR}) = 10^6 \log_2 (1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}}$$

Example 5 (continued)

The Shannon formula gives us 6 Mbps, the upper limit. For better performance we choose something lower, 4 Mbps, for example. Then we use the Nyquist formula to find the number of signal levels.

$$\boxed{4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L \quad \Rightarrow \quad L = 4}$$

Note

The Shannon capacity gives us the upper limit; the Nyquist formula tells us how many signal levels we need.

Thank You!